John D. Martin

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Education

2015 – 2021 **Ph.D. Mechanical Engineering**, Stevens Institute of Technology.
Advisor: *Brendan Englot*
Thesis: *Reinforcement Learning Algorithms for Representing and Managing Uncertainty in Robotics.*

2013 – 2015 **Graduate studies in Computer Science**, Columbia University.

2009 – 2012 **B.S. Physics, B.S. Aerospace Engineering**, University of Maryland.

Employment History

2024 – · · · · **Research Fellow,** Openmind Research Institute.
I conduct fundamental research in artificial intelligence with a focus on reinforcement learning. Specifically, I focus on how an agent can exploit its environment as an effective form of memory.

2023 – · · · · **Adjunct Professor,** University of Alberta, Department of Computing Science.
I supervise graduate students studying artificial intelligence and reinforcement learning. My students study algorithms for model-based RL in the continual setting.

2022 – 2024 **Research Scientist,** Intel Labs.
I performed applied and fundamental research. My applied work focused on the detection of CPU errors with code-specialized LLMs, fine-tuned for particular failure modalities. My fundamental work focused on sparsity in neural networks, model-based RL, and the Reward Hypothesis.

2021 – 2022 **Postdoctoral Fellow,** University of Alberta, Department of Computing Science.
Advisor: *Michael Bowling*
I studied the limits of reward as a mechanism to express goals, meta-learning algorithms for reinforcement learning, and algorithms for sample-efficient planning.

Summer 2020 **Research Scientist Intern,** DeepMind.
Advisor: *Joseph Modayil*
I studied RL algorithms for constructing sparse neural networks from unorganized observations.

2019 – 2020 **Student Researcher / Research Scientist Intern,** Google Brain.
Advisor: *Marc G. Bellemare*
I studied algorithms for detecting catastrophic interference in deep neural networks.

2017 – 2019 **Technical Consultant,** Piasecki Aircraft.
Focus areas: Conceptual design of experimental aircraft, proposal writing.

2012 – 2015 **Robotics and Flight Controls Engineer,** Sikorsky Aircraft.
Focus areas: Design of motion planning and control algorithms, automation of full-scale S-76.

Publications

Articles

1. **J. D. Martin**, F. Mince, E. Saleh, and A. Pajak, "Artifacts as Memory Beyond the Agent Boundary," *arXiv preprint arXiv:2604.08756*, 2026.
2. E. S. Lupu, P. Spieler, K. Javed, *et al.*, "The Open Ant: A Robot Platform for Reinforcement Learning Research," *Under Review*, 2026.
3. F. Dave-Louis, **J. D. Martin**, and M. Bowling, "On the Interplay Between Sparsity and Training in Deep Reinforcement Learning," *arXiv preprint arXiv:2501.16729*, 2025.
4. **J. D. Martin**, "The Methodological Tangle of AI Research," *RLDM (Oral)*, 2025.
5. **J. D. Martin**, B. Burega, L. Kapeluck, and M. Bowling, "Meta-Gradient Search Control: A Method for Improving the Efficiency of Dyna-style Planning," *arXiv preprint arXiv:2406.19561*, 2024.

6. R. Rafailov, K. B. Hatch, V. Kolev, **J. D. Martin**, M. Phielipp, and C. Finn, "MOTO: Offline Pre-training to Online Fine-tuning for Model-based Robot Learning," in *7th Annual Conference on Robot Learning*, 2023.
7. **J. D. Martin**, M. Bowling, D. Abel, and W. Dabney, "Settling the Reward Hypothesis," in *International Conference on Machine Learning*, PMLR, 2023.
8. B. Burega, **J. D. Martin**, and M. Bowling, "Learning to Prioritize Planning Updates in Model-based Reinforcement Learning," *NeurIPS Workshop on Meta Learning*, 2022.
9. E. Saleh, **J. D. Martin**, A. Koop, A. Pourzarabi, and M. Bowling, "Should Models Be Accurate?" *arXiv preprint arXiv:2205.10736*, 2022.
10. **J. D. Martin**, P. Szenher, X. Lin, and B. Englot, "The Stochastic Road Network Environment for Robust Reinforcement Learning," *ICRA Workshop on Releasing Robots into the Wild*, 2022.
11. **J. D. Martin**, "Time to Take Embodiment Seriously," *RLDM RL as Agency Workshop (Oral)*, 2022.
12. **J. D. Martin** and J. Modayil, "Adapting the Function Approximation Architecture in Online Reinforcement Learning," *arXiv preprint arXiv:2106.09776*, 2021.
13. **J. D. Martin**, "Reinforcement Learning Algorithms for Representing and Managing Uncertainty in Robotics," Ph.D. dissertation, Stevens Institute of Technology, Hoboken, New Jersey, USA, 2021.
14. F. Chen, **J. D. Martin**, Y. Huang, J. Wang, and B. Englot, "Autonomous exploration under uncertainty via deep reinforcement learning on graphs," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2020, pp. 6140–6147.
15. J. McConnell, **J. D. Martin**, and B. Englot, "Fusing concurrent orthogonal wide-aperture sonar images for dense underwater 3D reconstruction," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2020, pp. 1653–1660.
16. W. Fedus, D. Ghosh, **J. D. Martin**, M. G. Bellemare, Y. Bengio, and H. Larochelle, "On Catastrophic Interference in Atari 2600 Games," *arXiv preprint arXiv:2002.12499*, 2020.
17. **J. D. Martin**, M. Lyskawinski, X. Li, and B. Englot, "Stochastically Dominant Distributional Reinforcement Learning," in *International Conference on Machine Learning*, PMLR, 2020, pp. 6745–6754.
18. **J. D. Martin**, K. Doherty, C. Cyr, B. Englot, and J. Leonard, "Variational filtering with copula models for SLAM," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2020, pp. 5066–5073.
19. **J. D. Martin**, J. Wang, and B. Englot, "Sparse Gaussian Process Temporal Difference Learning for Marine Robot Navigation," in *Conference on Robot Learning*, PMLR, 2018, pp. 179–189.
20. **J. D. Martin** and B. Englot, "Extending model-based policy gradients for robots in heteroscedastic environments," in *Conference on Robot Learning*, PMLR, 2017, pp. 438–447.

Invited Talks

2026 **Artifacts as Memory Beyond the Agent Boundary**.
University of Massachusetts, Amherst
Institute of Philosophy, School of Advanced Studies, University of London
Noah's Ark Lab, Toronto
Alberta Machine Intelligence Institute, Edmonton
UTAR, Kuala Lumpur, Malaysia

2025 **AI for the Mind: The Quest Ever Thus**.
California Institute of Technology
NASA Jet Propulsion Laboratory

2024 **The Methodological Tangle of AI Research**.
Trinity College (RLDM), Dublin
University of Alberta, Edmonton
Google DeepMind, Seattle

Invited Talks (continued)

- **Reinforcement Learning and The Extended Mind Hypothesis**.
  Cohere for AI virtual talk
- 2023 **The Issaquah Plan**.
  Google DeepMind, Seattle
- 2022 **Learning to Prioritize Planning Updates in Model-based Reinforcement Learning**.
  University of Massachusetts, Amherst
- 2021 **Adapting the Function Approximation Architecture in Online Reinforcement Learning**.
  Google AI, Sparsity Reading Group
- 2020 **Uncertainty, Perception, and Their Lessons for Creating General-purpose Robots**.
  University of California, Berkeley
- 2019 **From Tasks to Timescales: A path to generalization in reinforcement learning**.
  Massachusetts Institute of Technology
  DeepMind, Edmonton
  Google Robotics, New York

Academic Service

Masters Thesis Advising

- 2025 **Haruto Tanaka**, University of Alberta, thesis committee member.
- 2024 – **Deepak Ranganatha Sastry Mamillapalli**, University of Alberta, co-advised with Matt Taylor.
- 2023 – **Luke Kapeluck**, University of Alberta, co-advised with Michael Bowling.
- 2022 – 2023 **Bradley Burega**, University of Alberta, co-advised with Michael Bowling.
- 2021 – 2024 **Fatima Davelouis**, University of Alberta, co-advised with Michael Bowling.

Organizer

- 2026 **Reinforcement Learning and Philosophy Workshop**:
- 2025 **Saving the Phenomena of Minds**: An RLDM workshop for examining RL as an computational theory of mental phenomena.
- 2024 – 2026 **Finding the Frame Workshop**: An RLC workshop for examining conceptual frameworks in RL.
- 2023 – **Seattle Minds and Machines Meetup**: a seminar series for Reinforcement Learning in Computer Science and Computational Neuroscience researchers in the Seattle-area.

Workflow Chair

- 2022 **AAAI**.

Program Chair

- 2026 **Openmind Winter School on AI and RL**.
- 2023 **Barbados RL Workshop**.
- 2021 **NAAMII Winter AI School**.
- 2020 **ICML RL Social**.

Reviewer

- 2026 **ICML, RLC, Workshops**.
- 2025 **ICML, RLC, RLDM, TMLR, CogSci, AAAI, Workshops**.
- 2024 **ICML, TMLR, Workshops**.
- 2023 **ICML, TMLR, Nature Machine Intelligence**.

Academic Service (continued)

2022 **ICML, NeurIPS, RLDM**.
2021 **ICML, NeurIPS, ICLR**.
2020 **ICML, NeurIPS, ICRA, WAFR, Workshops**.

Mentor

2023 **Cohere Labs Community**.
2022 **Neuromatch Academy**.
2020 **NeurIPS New in ML Workshop**.

Teaching Experience

Primary Instructor

Winter 2021 **RL Lecture Series,** Nepal Applied Mathematics and Informatics Institute.

Guest Lecturer

2017, 2020, 2021 **Advanced Robotics,** Stevens Institute of Technology.

Skills

Coding Python, LaTeX, C, C++, R, OCaml, …
Libraries JAX, Haiku,NumPy, Tensorflow, Pandas, Docker, Kubernetes, ROS, …

Miscellaneous Experience

Awards and Achievements

2019 – 2020 **Robert Brooks Stanley Doctoral Fellow**, Two-time recipient.
2015 **Department of Homeland Security Doctoral Fellow**.
Howard Hughes Award, American Helicopter Society.

References

Available on Request